

Periodic solutions for nonlinear differential equations in Banach spaces with generalized variable-exponent operators

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Motivation and the main problem

Periodic models in ordinary differential equations

Many systems (physical, biological, among others) are subject to recurrent effects, such as periodic forcing, cyclic loading, or oscillatory behaviours.

Usually, a simple way of modeling these phenomena is by second-order ordinary differential systems

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Unlike IVP, the initial state is not prescribed: it must be chosen so that the trajectory closes after one period. Thus, periodic problems are difficult and have been a classical motivation for several methods of nonlinear analysis, including topological degree and continuation techniques.

From classical equations to nonlinear constitutive laws

The previous model assumes a linear response with respect to the velocity, since the principal term is u'' .

In nonlinear models, this response may instead be described by a constitutive law ϕ , leading to

$$\frac{d}{dt}\phi(u'(t)) = f(t, u(t), u'(t)).$$

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More generally, the constitutive response may also vary with time. For instance, a time-dependent coefficient may lead to

$$\frac{d}{dt}\left[a(t)\phi(u'(t))\right] = f(t, u(t), u'(t)),$$

Previous theory in finite dimensions

Both time-independent and time-dependent constitutive laws can be conveniently included in periodic problems of the form

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = f(t, u(t), u'(t)), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases} \quad u : [0, T] \rightarrow \mathbb{R}^N.$$

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This naturally raises the question of extending this theory to infinite-dimensional state spaces.

From $\mathbb{R}^N$ to a Banach space

Such systems arise, for instance, in spatially distributed models, where the state at each time is itself a function:

$$u(t) = u(t, \cdot).$$

Therefore, the natural state space is no longer $\mathbb{R}^N$, but a Banach space X .

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What is gained

The abstract equation can describe problems related to

- ▶ nonlinear PDEs;
- ▶ abstract differential equations;
- ▶ evolution problems.

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What becomes more difficult

In infinite dimension:

- ▶ bounded sets need not be relatively compact;
- ▶ Some topological methods change drastically, or simply cease to exist.

The periodic problem studied in the thesis

Let X be a real separable reflexive Banach space, with dual X^* . We study

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = f(t, u(t)), & t \in (0, T), \\ u(0) = u(T), & u'(0) = u'(T), \end{cases} \quad (\text{P})$$

where

$$S : [0, T] \times X \rightarrow X^*, \quad f : [0, T] \times X \rightarrow X^*.$$

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Definition of solution

A solution of (P) is a function $u \in C^1([0, T]; X)$ such that

$$S(\cdot, u'(\cdot)) \in W^{1,1}([0, T]; X^*)$$

and the differential equation holds a.e. on $(0, T)$.

Basic notation and periodic spaces

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- For a Banach space Y , we write

$$C_T([0, T]; Y) := \{u \in C([0, T]; Y) : u(0) = u(T)\},$$

and

$$C_0([0, T]; Y) := \{v \in C([0, T]; Y) : v(0) = v(T) = 0\}.$$

Structural assumptions on S

We assume that $S : [0, T] \times X \rightarrow X^*$ satisfies:

(S0) Continuity, normalization, and periodicity. The map S is continuous and, for every $t \in [0, T]$ and $x \in X$,

$$S(t, x) = 0 \iff x = 0, \quad S(0, x) = S(T, x).$$

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(S1a) Local uniform strict monotonicity. For every $M > 0$, there exists a continuous non-decreasing function

$$\omega_M : [0, \infty) \rightarrow [0, \infty), \quad \omega_M(0) = 0, \quad \omega_M(r) > 0 \text{ for } r > 0,$$

such that, for every $t \in [0, T]$ and $\|x\|_X, \|y\|_X \leq M$,

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(S1b) Uniform coercivity. There exists a continuous non-decreasing function

$$\alpha : [0, \infty) \rightarrow [0, \infty), \quad \alpha(r) \rightarrow \infty \text{ as } r \rightarrow \infty,$$

such that, for every $t \in [0, T]$ and $x \in X$,

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(S2) Boundedness on bounded sets. There exists a continuous function $\beta : [0, T] \times [0, \infty) \rightarrow [0, \infty)$ such that, for every $t \in [0, T]$ and $x \in X$,

$$\|S(t, x)\|_{X^*} \leq \beta(t, \|x\|_X).$$

An example: the $p(t)$ -Laplacian-type operator

Let H be a real Hilbert space and let $\mathcal{R} : H \rightarrow H^*$ be the Riesz isomorphism.

Assume that $p \in C([0, T])$ satisfies

$$1 < p_- \leq p(t) \leq p_+ < \infty, \quad p(0) = p(T).$$

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Variable-exponent operator

Define

$$S(t, 0) := 0, \quad S(t, x) := \|x\|_H^{p(t)-2} \mathcal{R}x, \quad x \neq 0.$$

Then

$$S : [0, T] \times H \rightarrow H^*$$

satisfies **(S0)**–**(S2)**.

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In the Hilbert-space formulation, we may work directly with

$$S(t, x) = \|x\|_H^{p(t)-2} x, \quad S : [0, T] \times H \rightarrow H.$$

Invertibility and properties of the inverse

Fix $t \in [0, T]$ and set

$$S_t := S(t, \cdot) : X \rightarrow X^*.$$

Under **(S0)**–**(S2)**, S_t is continuous, strictly monotone, and coercive. Hence, by the Minty–Browder theorem, S_t is bijective.

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Properties of Γ

- ▶ $\Gamma : [0, T] \times X^* \rightarrow X$ is jointly continuous and strictly monotone;
- ▶ $\|\Gamma(t, y)\|_X \rightarrow \infty$ as $\|y\|_{X^*} \rightarrow \infty$, uniformly in t ;
- ▶ if $g_n \rightarrow g$ in $C([0, T]; X^*)$, then

$$\Gamma(\cdot, g_n(\cdot)) \rightarrow \Gamma(\cdot, g(\cdot)) \quad \text{in } C([0, T]; X).$$

Nonlinear Green operator and generalized reduction

Compatibility and first integration

Let u solve the *non-homogeneous periodic problem*,

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = h(t), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases} \quad h \in L^1([0, T]; X^*). \quad (\text{NH})$$

Define

$$Qh := \int_0^T h(t) dt, \quad H(h)(t) := \int_0^t h(s) ds.$$

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$$u(t) = u(0) + \int_0^t \Gamma(s, c + H(h)(s)) ds, \quad c \in X^*$$

Position periodicity and the implicit equation

Evaluating the representation formula at $t = T$, we obtain

$$u(T) - u(0) = \int_0^T \Gamma(t, c + H(h)(t)) dt.$$

Therefore,

$$u(T) = u(0) \iff \int_0^T \Gamma(t, c + H(h)(t)) dt = 0.$$

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To express this condition, for each $g \in C([0, T]; X^*)$, define

$$\Phi_g : X^* \rightarrow X, \quad \Phi_g(c) := \int_0^T \Gamma(t, c + g(t)) dt.$$

Hence, the periodicity condition is equivalent to requiring the integration constant $c \in X^*$ to solve

$$\Phi_{H(h)}(c) = 0.$$

Solvability of the implicit equation

Key lemma

For every $g \in C([0, T]; X^*)$, there exists a unique $\hat{c}(g) \in X^*$ such that

$$\Phi_g(\hat{c}(g)) = 0.$$

Moreover, $\hat{c}: C([0, T]; X^*) \rightarrow X^*$ is continuous and maps bounded sets into bounded sets.

The nonlinear Green operator

From the previous construction, the integration constant is $c = \widehat{c}(H(h))$. This leads us to define

$$\mathcal{T}(h)(t) := \int_0^t \Gamma\left(s, \widehat{c}(H(h)) + H(h)(s)\right) ds.$$

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Since $\widehat{c}(H(h))$ solves $\Phi_{H(h)}(c) = 0$, we have

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Therefore $\mathcal{T} : L^1([0, T]; X^*) \longrightarrow C_0([0, T]; X)$.

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Solvability and representation

The problem (NH) has a solution if and only if there exists $u \in C_T([0, T]; X)$ such that

$$\begin{cases} u(t) = u(0) + \mathcal{T}(h)(t), & t \in [0, T] \\ Qh = 0. \end{cases}$$

From the main problem to an abstract system

We now return to the nonlinear periodic problem

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T). \end{cases}$$

Assume that f is Carathéodory and satisfies the local growth conditions needed to define the Nemytskii operator $N_f : C([0, T]; X) \rightarrow L^1([0, T]; X^*)$, $N_f(u)(t) := f(t, u(t))$.

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The compactness defect

In infinite dimension, $u \mapsto u(0)$ is not a completely continuous operator. Hence, the constant component $u(0)$ must be separated.

The generalized Lyapunov–Schmidt decomposition

For every $u \in C_T([0, T]; X)$, we have the following unique decomposition,

$$u = u(0) + (u - u(0)), \quad u(0) \in X, \quad u - u(0) \in C_0([0, T]; X).$$

Since $X \cap C_0([0, T]; X) = \{0\}$ (using X represented as the constant functions),

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Set

$$a := u(0), \quad v := u - a.$$

Substituting $u = a + v$ into the abstract system and cancelling the constant component a , we obtain

$$\boxed{\begin{cases} v = \mathcal{T}(N_f(a + v)), \\ QN_f(a + v) = 0, \end{cases} \quad a \in X, \quad v \in C_0([0, T]; X).} \quad (\text{LS})$$

Two existence strategies

The two methods developed in this thesis differ in how they treat the averaged constraint

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Second method: approximate the coupled system

When the constraint cannot be solved explicitly for a , we solve finite-dimensional approximations of the full system and then pass to the limit to recover a solution of the original problem.

First existence method: mean-admissible nonlinearities

Mean-admissibility and the averaged constraint

Let Y, Z be reflexive Banach spaces and $f : [0, T] \times Y \rightarrow Z$. For $v \in C_0([0, T]; Y)$, define

$$F_v(a) := \int_0^T f(t, a + v(t)) dt.$$

Definition

The map f is *mean-admissible* if, for every v ,

$$F_v(a) = 0$$

has a unique solution $a = \hat{A}_f(v)$, where $\hat{A}_f : C_0([0, T]; Y) \rightarrow Y$ is continuous and maps bounded sets into bounded sets.

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In the generalized Lyapunov–Schmidt system, $QN_f(a + v) = F_v(a)$. Hence,

$$\boxed{QN_f(a + v) = 0 \iff a = \hat{A}_f(v).}$$

A sufficient criterion for mean-admissibility

The implicit equation for Γ suggests imposing analogous monotonicity and coercivity assumptions on f .

Proposition

Let Y be a real separable reflexive Banach space and let $f : [0, T] \times Y \rightarrow Y^*$ be continuous. Assume, uniformly in $t \in [0, T]$, that:

- (i) There exists a non-decreasing function $\eta : [0, \infty) \rightarrow [0, \infty)$, with $\eta(r) > 0$ for $r > 0$, such that

$$\langle f(t, x) - f(t, y), x - y \rangle \geq \eta(\|x - y\|_Y).$$

- (ii) There exist $\Theta : [0, \infty) \rightarrow \mathbb{R}$ and a non-decreasing function $\gamma : [0, \infty) \rightarrow [0, \infty)$ such that

$$\langle f(t, x), x \rangle \geq \Theta(\|x\|_Y) \|x\|_Y, \quad \|f(t, x)\|_{Y^*} \leq \gamma(\|x\|_Y),$$

and

$$\Theta(r) \rightarrow \infty, \quad \Theta(r)r - M\gamma(r) \rightarrow \infty \quad \text{for every } M > 0.$$

Then f is *mean-admissible*.

The criterion is sufficient, but not necessary

Let $T = 2\pi$, let H be a real Hilbert space, and consider

$$f(t, x) = m(t)x + b(t), \quad m(t) := 1 + 2 \sin t.$$

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The criterion is sufficient, but not necessary

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However, the averaged equation

$$\int_0^{2\pi} f(t, a + v(t)) dt = 0$$

has the unique solution

$$\hat{A}_f(v) = -\frac{1}{2\pi} \left(\int_0^{2\pi} m(t)v(t) dt + \int_0^{2\pi} b(t) dt \right).$$

Fixed-point reduction and the compactness issue

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To apply topological methods such as Leray–Schauder degree, the right-hand side must define a completely continuous operator.

However, for an arbitrary Banach space X ,

$$\mathcal{T} : L^1([0, T]; X^*) \rightarrow C_0([0, T]; X)$$

need not be completely continuous.

Recovering compactness through a Gelfand triple

To recover compactness, we assume that X is part of a compact Gelfand triple

$$X \xhookrightarrow{i} H \cong H^* \xhookrightarrow{i^*} X^*,$$

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H -valued periodic solutions

Let $f : [0, T] \times H \rightarrow H$. An H -valued periodic solution is a function

$$u = a + v \in C_T([0, T]; H), \quad a \in H, \quad v \in C_0([0, T]; X) \cap C^1([0, T]; X),$$

satisfying

$$\begin{cases} \frac{d}{dt} S(t, u'(t)) = i^* f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T). \end{cases}$$

The compact fixed-point problem

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Consequently, the reduced equation in the Gelfand setting is

$$v = \mathcal{T} \left(i^* N_f \left(\widehat{A}_f(v) + v \right) \right), \quad v \in C_0([0, T]; H). \quad (\text{FP}_H)$$

Its right-hand side is completely continuous, so Leray–Schauder degree can be applied.

A continuation theorem

Theorem

Let $\Omega \subset C_T([0, T]; H)$ be bounded and open, and let $f : [0, T] \times H \rightarrow H$ be a *mean-admissible* Carathéodory function. Assume that:

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$$\begin{cases} \frac{d}{dt} S(t, u'(t)) = \lambda i^* f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases}$$

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Then (P) admits an H -valued periodic solution $u \in \overline{\Omega}$.

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- ▶ Transfer the domain through R :

$$\Omega_0 := R^{-1}(\Omega \cap M).$$

Then Ω_0 is bounded and open. Moreover,

$$\widehat{A}_f(0) = c \quad \implies \quad 0 \in \Omega_0.$$

Proof idea II: compact homotopy and degree

- For $\lambda \in [0, 1]$, define

$$\mathcal{G}(\lambda, v) := \mathcal{T}\left(\lambda i^* N_f(R(v))\right).$$

The compactness of $\mathcal{T}$ implies that $\mathcal{G}$ is completely continuous.

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$$d_{LS}[I - \mathcal{G}(1, \cdot), \Omega_0, 0] = d_{LS}[I, \Omega_0, 0] = 1.$$

- Since the degree is different from zero, there exists $v \in \Omega_0$ with $v = \mathcal{G}(1, v)$. Therefore

$$u := R(v) = \widehat{A}_f(v) + v$$

is an H -valued periodic solution with $u \in \Omega$.

Application: a double-phase periodic problem

Let $\mathcal{O} \subset \mathbb{R}^d$ be open, bounded with Lipschitz boundary, and set

$$X = W_0^{1,\beta}(\mathcal{O}) \hookrightarrow H = L^2(\mathcal{O}) \hookrightarrow X^* = W^{-1,\beta'}(\mathcal{O}), \quad \beta > 2.$$

Let $p \in C([0, 2\pi])$ such that

$$1 < p_- \leq p(t) \leq p_+ \leq \beta, \quad p(0) = p(2\pi),$$

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All the assumptions of the continuation theorem can be verified. Therefore, the problem admits an H -valued periodic solution.

Second existence method: Hartman–Galerkin

From monotonicity to the Galerkin strategy

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In infinite dimension, a natural strategy is to approximate each problem by finite-dimensional problems, solve the corresponding projected problems and verify that this sequence converges to a solution of the original problem. This is the Galerkin method.

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finite-dimensional approximation $\longrightarrow$ solutions u_N $\longrightarrow$ uniform estimates $\longrightarrow$ limit solution

Variable-exponent functional setting

Let $p \in C([0, T])$ satisfy

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(H0) Continuity, normalization, and periodicity. The map S is continuous and satisfies

$$S(t, x) = 0 \iff x = 0, \quad S(0, x) = S(T, x),$$

for every $t \in [0, T]$ and $x \in X$.

Stronger hypotheses on the principal operator

(H1) Local uniform strict monotonicity. For every $M > 0$, there exists a continuous non-decreasing function

$$\omega_M : [0, \infty) \rightarrow [0, \infty), \quad \omega_M(0) = 0, \quad \omega_M(r) > 0 \text{ for } r > 0,$$

such that

$$\langle S(t, x) - S(t, y), x - y \rangle \geq \omega_M(\|x - y\|_X)$$

for every $t \in [0, T]$ and $\|x\|_X, \|y\|_X \leq M$.

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(H2) Variable-exponent coercivity. There exists $c_0 > 0$ such that

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(H3) Variable-exponent growth. There exists $c_1 > 0$ such that

$$\|S(t, x)\|_{X^*} \leq c_1 \left(1 + \|x\|_X^{p(t)-1}\right)$$

for every $t \in [0, T]$ and $x \in X$.

Assumptions on the nonlinearity

Let $m, k : [0, T] \times X \rightarrow X^*$ be Carathéodory maps and set

$$f(t, x) := m(t, x) + k(t, x).$$

(F1) Local growth. For every $r > 0$, there exists $\eta_r \in L^{p'(\cdot)}([0, T])$ such that, for a.e. $t \in [0, T]$,

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(F3_e) Hartman condition. There exists $R > 0$ such that, for a.e. $t \in [0, T]$,

$$\langle f(t, x), x \rangle \geq 0 \quad \text{whenever } \|x\|_X \geq R.$$

Main Hartman-type theorem

Theorem

Assume that S satisfies **(H0)–(H3)**, and that

$$f(t, x) = m(t, x) + k(t, x)$$

satisfies **(F1)**, **(F2_m)**, **(F2_{ws})**, and **(F3_e)**.

Then (P) admits at least one periodic solution $u \in C^1([0, T]; X)$.

The finite-dimensional framework

Since X is separable, choose a linearly independent sequence

$$(e_j)_{j \geq 1} \subset X, \quad \overline{\text{span}\{e_j : j \geq 1\}}^X = X,$$

and set

$$X_N := \text{span}\{e_1, \dots, e_N\}.$$

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Endow X_N with the auxiliary inner product $(\cdot, \cdot)_N$ for which $(e_1, \dots, e_N)$ is orthonormal, and define

$$R_N y := \sum_{j=1}^N \langle y, e_j \rangle e_j, \quad y \in X^*,$$

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Projected S operator

For every N and t , $S_N(t, \cdot) : X_N \rightarrow X_N$ is a homeomorphism. Moreover, $\Gamma_N(t, \cdot) := (S_N(t, \cdot))^{-1}$ verifies all the properties of Γ .

The regularized projected problems

To obtain a strict exterior inequality while recovering the original problem as $N \rightarrow \infty$, set

$$\varepsilon_N := \frac{1}{N}, \quad \tilde{f}_N(t, x) := f(t, x) + \varepsilon_N S(t, x).$$

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Hence, we will consider, for $\lambda \in [0, 1]$, the one-parameter family of problems,

$$\begin{cases} \frac{d}{dt} S_N(t, u'_N(t)) = \lambda R_N \tilde{f}_N(t, u_N(t)), \\ u_N(0) = u_N(T), \quad u'_N(0) = u'_N(T). \end{cases} \quad (\text{P}_{N,\lambda})$$

Finding the projected solutions by degree theory

Write

$$u_N = a_N + v_N, \quad a_N = u_N(0), \quad v_N = u_N - a_N,$$

hence $a_N \in X_N$, $v_N \in E_N := C_0([0, T]; X_N)$, where $\|v\|_{E_N} := \max_{t \in [0, T]} \|v(t)\|_X$.

Finding the projected solutions by degree theory

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Define the associated fixed-point operator $\mathcal{K}_{N, \lambda} : X_N \times E_N \rightarrow X_N \times E_N$ as:

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This operator is completely continuous since X_N is finite-dimensional (standard argument using the properties of Γ_N and $\mathcal{Q}_N$).

Proof idea I: uniform Hartman bounds

- Let u_N solve $(P_{N,\lambda})$. Testing the equation with u_N gives

$$c_0 \int_0^T \|u'_N\|_X^{p(t)} dt \leq -\lambda \int_0^T \langle \tilde{f}_N(t, u_N), u_N \rangle dt.$$

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$$\|u_N\|_{C([0,T];X)} \leq C \quad \implies \quad \|u_N\|_V \leq C'.$$

Proof idea II: degree and projected solutions

The uniform estimate for u_N over $C([0, T]; X)$ gives

$$\|a_N\|_X + \|v_N\|_{E_N} \leq C_1.$$

Choose $\rho > \max\{C_1, R\}$, and define the following open bounded set

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- At $\lambda = 0$ the fixed point problem is $(a, v) = (a - \mathcal{Q}_N(a + v), 0)$; hence, the problem is equivalent to $\mathcal{Q}_N(a) = 0$. By an auxiliary homotopy and the exterior inequality, we get

$$d_{LS}[I - \mathcal{K}_{N,1}, \Omega_N, 0] = d_{LS}[I - \mathcal{K}_{N,0}, \Omega_N, 0] = d_B[\mathcal{Q}_N, B_\rho^{X_N}, 0] = 1.$$

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- ▶ Hence $\mathcal{K}_{N,1}$ has a fixed point, and the regularized projected problem admits a solution u_N .

Proof idea III: passage to the limit

- Since $\|u_N\|_V \leq C$ uniformly, and V is reflexive, up to a subsequence,

$$u_N \rightharpoonup u \quad \text{in } V, \quad u_N(t) \rightharpoonup u(t) \quad \text{in } X.$$

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$$k(\cdot, u_N) \rightarrow k(\cdot, u) \quad \text{in } L^{p'(\cdot)}([0, T]; X^*), \quad \varepsilon_N S(\cdot, u_N) \rightarrow 0.$$

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- Using the monotonicity of S and m , Minty's trick identifies the weak limit and yields

$$\int_0^T \langle S(t, u'), \varphi' \rangle dt + \int_0^T \langle f(t, u), \varphi \rangle dt = 0, \quad \forall \varphi \in V.$$

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- ▶ Thus, u is a weak periodic solution. Setting $q := S(\cdot, u')$, we have

$$q \in W^{1, p'(\cdot)}([0, T]; X^*), \quad u'(t) = \Gamma(t, q(t)) \in C([0, T]; X),$$

and therefore $u \in C^1([0, T]; X)$. The periodicity of u follows from the same equation.

Two limiting cases

Purely monotone nonlinearity

Assume that S satisfies **(H0)–(H3)**, and that $f : [0, T] \times X \rightarrow X^*$ satisfies **(F1)** and is monotone in its second variable. If, for some $R > 0$,

$$\langle f(t, x), x \rangle \geq 0 \quad \text{for a.e. } t \in [0, T] \text{ and } \|x\|_X = R,$$

then the periodic problem admits a solution. Any two solutions differ by a constant element of X ; if $f(t, \cdot)$ is strictly monotone, the solution is unique.

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Purely weak-to-strong nonlinearity

Suppose that X is infinite-dimensional. If f satisfies **(F1)**, is sequentially weak-to-strong continuous, and

$$\langle f(t, x), x \rangle \geq 0 \quad \text{for } \|x\|_X \geq R,$$

then every periodic solution is stationary. In particular, $u \equiv 0$ is a solution.

Radial Hilbert-space operators

Let X be a Hilbert space and assume that S satisfies **(H0)–(H3)** and has the radial form

$$S(t, 0) = 0, \quad S(t, x) = \beta(t, \|x\|_X) \mathcal{R}x, \quad \beta(t, r) > 0.$$

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Proposition

Under the hypotheses of the main Hartman-type theorem, the periodic problem admits a solution u satisfying

$$\|u(t)\|_X \leq R, \quad t \in [0, T].$$

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This partially extends the corresponding finite-dimensional radial Hartman bound of [8] to the Hilbert-space setting.

Application: a nonlocal Sobolev-type problem

Let $\mathcal{O} \subset \mathbb{R}^d$ be defined as the previous application, $X = H_0^1(\mathcal{O})$, $X^* = H^{-1}(\mathcal{O})$, $A = -\Delta$ (in weak form), and

$$S(t, x) = \|x\|_X^{p(t)-2} Ax.$$

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We study

$$\begin{cases} \partial_t \left[-\|\nabla u_t(t, \cdot)\|_2^{p(t)-2} \Delta_x u_t(t, x) \right] = -\frac{\Delta_x u(t, x)}{\sqrt{1 + \|\nabla u(t, \cdot)\|_2^2}} + b(t), \\ u|_{\partial\mathcal{O}}(t, \cdot) = 0, \quad u(0) = u(T), \quad u_t(0) = u_t(T). \end{cases}$$

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Then there exists a unique nonstationary periodic solution satisfying

$$\|\nabla u(t, \cdot)\|_{L^2(\mathcal{O})} \leq \frac{B}{\sqrt{1 - B^2}}, \quad t \in [0, T].$$

Concluding remarks

Conclusions

- We introduced a broad class of monotone operators S , including p -Laplacian, $p(t)$ -Laplacian, and double-phase type operators.

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- ▶ The results cover many types of problems, involving different nonlinearities and differential operators.

Open problems and future directions

Derivative-dependent nonlinearities

Can the existence methods developed here be extended to

$$\frac{d}{dt}S(t, u'(t)) = f(t, u(t), u'(t))?$$

This would require new compactness arguments, a priori estimates for u' , and possibly additional structural conditions on f .

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Further applications of the decomposition

Can the decomposition underlying the generalized Lyapunov–Schmidt system be adapted to other problems with a noncompact or infinite-dimensional constant component?

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Thank you

Questions

Periodic solutions for nonlinear differential equations in Banach spaces with generalized variable-exponent operators

Jorge Hernán Zacarías Novoa Contreras

Supervisor: Raúl Manásevich Tolosa



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Santiago de Chile, 2026

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Appendix

Can the theory cover the classical wave equation?

Consider

$$\begin{cases} u_{tt}(t, x) = u_{xx}(t, x), & (t, x) \in [0, T] \times (0, 1), \\ u(t, 0) = u(t, 1) = 0, & t \in [0, T], \\ u(0, \cdot) = u(T, \cdot), \quad u_t(0, \cdot) = u_t(T, \cdot). \end{cases}$$

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If $H = L^2(0, 1)$, we may write formally

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Thus $S : H \rightarrow H$ satisfies **(S0)**–**(S2)**.

However,

$$u \longmapsto u_{xx}$$

is not defined on all of $L^2(0, 1)$, since

$$D(\partial_{xx}) = H^2(0, 1) \cap H_0^1(0, 1).$$

Hence the L^2 -realization is not directly covered.

The obstruction in the energy space

Let

$$H = H_0^1(0, 1), \quad (u, v)_H := \int_0^1 u_x v_x \, dx,$$

and define $B : H \rightarrow H$ by

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Take

$$\psi_n(x) = \frac{\sqrt{2}}{n\pi} \sin(n\pi x), \quad \|\psi_n\|_H = 1.$$

Then

$$(B\psi_n, \psi_n)_H = \frac{1}{(n\pi)^2} \rightarrow 0.$$

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Hence B is not coercive in H , and the classical wave equation is not directly covered by the present framework.

The key lemma: existence and uniqueness

For $g \in C([0, T]; X^*)$, define

$$\Phi_g(c) := \int_0^T \Gamma(t, c + g(t)) dt, \quad \Psi_g := J_X \circ \Phi_g : X^* \rightarrow X^{**}.$$

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Continuity of Γ makes Ψ_g continuous, hence hemicontinuous. Moreover, Ψ_g is strictly monotone, since for $c_1 \neq c_2$,

$$\langle \Psi_g(c_1) - \Psi_g(c_2), c_1 - c_2 \rangle = \int_0^T \langle c_1 - c_2, \Gamma(t, c_1 + g(t)) - \Gamma(t, c_2 + g(t)) \rangle dt > 0.$$

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Set $z_c(t) := \Gamma(t, c + g(t))$. Coercivity of S gives

$$\langle \Psi_g(c), c \rangle \geq \int_0^T [\alpha(\|z_c(t)\|_X) - \|g\|_\infty] \|z_c(t)\|_X dt.$$

As $\|c\|_{X^*} \rightarrow \infty$, $\|z_c(t)\|_X \rightarrow \infty$ uniformly in t ; hence

$$\langle \Psi_g(c), c \rangle \rightarrow +\infty.$$

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Since X^* is reflexive, the monotone Bolzano theorem [18, Theorem 4] yields a unique zero.

The key lemma: boundedness on bounded sets

Let $\Lambda \subset C([0, T]; X^*)$ be bounded and choose $M > 0$ such that

$$\|g\|_{\infty} \leq M \quad \text{for every } g \in \Lambda.$$

Write

$$c := \widehat{c}(g), \quad z(t) := \Gamma(t, c + g(t)).$$

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If $\|c\|_{X^*}$ were arbitrarily large, then $\|z(t)\|_X$ would be uniformly large, so that

$$\alpha(\|z(t)\|_X) \geq 2M.$$

Consequently,

$$2M \int_0^T \|z(t)\|_X dt \leq \int_0^T \langle c + g(t), z(t) \rangle dt \leq M \int_0^T \|z(t)\|_X dt,$$

a contradiction. Therefore $\widehat{c}: C([0, T]; X^*) \rightarrow X^*$ maps bounded sets into bounded sets.

The key lemma: continuity of $\widehat{c}$

Let $g_n \rightarrow g$ in $C([0, T]; X^*)$, and set

$$c_n := \widehat{c}(g_n), \quad c := \widehat{c}(g), \quad \Delta_n(t) := \Gamma(t, c_n + g_n(t)) - \Gamma(t, c + g(t)).$$

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The preceding result makes (c_n) bounded; hence both values of Γ lie in a fixed ball $B_{M_B}^X$. Using

$$\Phi_{g_n}(c_n) = \Phi_g(c) = 0$$

and (S1), we obtain

$$\int_0^T \omega_{M_B}(\|\Delta_n(t)\|_X) dt \leq 2M_B T \|g_n - g\|_\infty \longrightarrow 0.$$

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Since $S(t_0, \Gamma(t_0, c_{n_k} + g_{n_k}(t_0))) = c_{n_k} + g_{n_k}(t_0)$, the continuity of $S(t_0, \cdot)$ and $g_{n_k}(t_0) \rightarrow g(t_0)$ imply

$$c_{n_k} \rightarrow c \quad \text{in } X^*.$$

Since every subsequence of c_n has a further subsequence that converges to c , this means that the entire sequence converges, hence $\widehat{c}$ is continuous.

Why does the Green operator become compact in H ?

Assume the compact Gelfand triple

$$X \hookrightarrow H \cong H^* \hookrightarrow X^*.$$

For a bounded sequence $(h_n) \subset L^1([0, T]; X^*)$, write

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The estimates for Γ give:

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Therefore, Arzelà–Ascoli yields

$$\mathcal{T} : L^1([0, T]; X^*) \longrightarrow C_0([0, T]; H) \text{ completely continuous.}$$

Why is the projected problem regularized?

In the Galerkin problems, set

$$\widetilde{f}_N(t, x) = f(t, x) + \varepsilon_N S(t, x), \quad \varepsilon_N = \frac{1}{N}.$$

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Thus, the regularization:

- ▶ gives the strict exterior inequality needed for the degree;
- ▶ excludes solutions on the boundary of the Galerkin domain;
- ▶ disappears in the limit, since

$$\varepsilon_N S(\cdot, u_N) \rightarrow 0.$$

Minty's trick I: where does the inequality come from?

Let $u_N \rightharpoonup u$ in V . For any $w \in V$, choose $w_N \in V_N$ such that $w_N \rightarrow w$ strongly in V .

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By monotonicity of $S(t, \cdot)$ and $m(t, \cdot)$,

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Using $w_N \rightarrow w$ in V , the weak-to-strong convergence of k , and the vanishing of the regularization, we obtain

$$\boxed{0 \geq \int_0^T \langle S(t, w'), u' - w' \rangle dt + \int_0^T \langle m(t, w) + k(t, u), u - w \rangle dt} \quad (w \in V).$$

Minty's trick II: recovering the limit equation

Starting from

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take $w = u - \lambda v$, with $v \in V$ and $\lambda > 0$. Dividing by λ gives

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Letting $\lambda \rightarrow 0^+$, continuity yields

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Replacing v by $-v$ gives the reverse inequality. Therefore

$$\boxed{\int_0^T \langle S(t, u'), v' \rangle dt + \int_0^T \langle f(t, u), v \rangle dt = 0 \quad \text{for every } v \in V.}$$

Thus u is a weak solution of the limit problem.

Why is the weak solution actually C^1 ?

After the passage to the limit, define

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Using the continuous inverse of $S(t, \cdot)$,

$$u'(t) = \Gamma(t, q(t)) \in C([0, T]; X).$$

Hence

$$u \in C^1([0, T]; X).$$

Can the mean-admissible method include u' ?

Consider

$$\frac{d}{dt}S(t, u'(t)) = f(t, u(t), u'(t)).$$

The reduced equation would involve

$$v = \mathcal{T}\left(N_f\left(\widehat{A}_f(v) + v\right)\right),$$

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Indeed,

$$(\mathcal{T}h)'(t) = \Gamma\left(t, \widehat{c}(H(h)) + H(h)(t)\right),$$

and the current estimates give uniform boundedness, but not the equicontinuity or relative compactness of this derivative family.

The first method therefore does not extend directly.

A possible Galerkin extension with dependence on u'

A possible route is to work in a Hilbert space H and consider

$$\frac{d}{dt}S(t, u') = f(t, u, u').$$

Inspired by classical Hilbert-space results, one could impose the joint monotonicity condition

$$\left(S(t, y) - S(t, z), y - z \right)_H + \left(f(t, x, y) - f(t, w, z), x - w \right)_H \geq 0$$

for every $x, w, y, z \in H$.

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When $S = I$, these terms become

$$\|y - z\|_H^2, \quad \|y\|_H^2 + \left(f(t, x, y), x \right)_H,$$

as in the classical second-order Hilbert-space setting shown in [17].

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$$\|y - z\|_H^2, \quad \|y\|_H^2 + \left(f(t, x, y), x \right)_H,$$

as in the classical second-order Hilbert-space setting shown in [17].

These conditions suggest a Galerkin–Minty extension, but a complete proof would require new growth estimates and a careful limit argument.

Why is the purely monotone case stronger?

Assume that $f(t, \cdot)$ is monotone and that

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$$y := \frac{R}{r}x.$$

Then $\|y\|_X = R$, and monotonicity gives

$$0 \leq \langle f(t, x) - f(t, y), x - y \rangle = \left(1 - \frac{R}{r}\right) \langle f(t, x) - f(t, y), x \rangle.$$

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If u, w are two solutions, testing the difference with $u - w$ gives

$$0 = \int_0^T \langle S(t, u') - S(t, w'), u' - w' \rangle dt + \int_0^T \langle f(t, u) - f(t, w), u - w \rangle dt.$$

Both terms are nonnegative, hence $u' = w'$, so $u - w$ is constant. Strict monotonicity of f gives $u = w$.

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Assume that X is infinite-dimensional and that $f(t, \cdot)$ is weak-to-strong continuous.

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Choose

$$e_j \rightharpoonup 0, \quad \|e_j\|_X = 1.$$

For fixed $x \in X$, take $L > R + \|x\|_X$. Then $\|x + Le_j\|_X > R$ so the Hartman condition gives

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Taking $x = \pm \varepsilon v$ and $\varepsilon \rightarrow 0$ gives $f(t, 0) = 0$. Finally, testing the equation with u ,

$$0 = \int_0^T \langle S(t, u'), u' \rangle dt + \int_0^T \langle f(t, u), u \rangle dt.$$

Both terms are nonnegative, hence $u' = 0$, and hence every solution is constant with respect to $t \in [0, T]$.

Radial case: proof idea

Set

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$$(S_N(t, u'_N), u_N)_X > 0.$$

But radiality gives

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Therefore

$$\|u_N(t)\|_X \leq R \implies \|u(t)\|_X \leq R.$$

Double-phase periodic problem

$$\begin{aligned} \frac{\partial}{\partial t} \left[-\operatorname{div} \left(|\nabla u_t|^{p(t)-2} \nabla u_t + |\nabla u_t|^{\beta-2} \nabla u_t \right) \right] &= f(t, u) \quad \text{in } (0, T) \times \Omega, \\ u(t, \cdot) &= 0 \quad \text{on } \partial\Omega, \\ u(0, \cdot) &= u(T, \cdot), \quad u_t(0, \cdot) = u_t(T, \cdot). \end{aligned}$$

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$$X = W_0^{1,\beta}(\Omega), \quad 1 < p_- \leq p(t) \leq p_+ \leq \beta, \quad \beta > 2.$$

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The spatial operator is understood weakly as $S(t, \cdot) : X \rightarrow X^*$, where

$$\langle S(t, v), \phi \rangle = \int_{\Omega} \left(|\nabla v|^{p(t)-2} \nabla v + |\nabla v|^{\beta-2} \nabla v \right) \cdot \nabla \phi \, dx, \quad \phi \in X.$$

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Thus the PDE has the abstract form

$$\frac{d}{dt} S(t, u_t(t)) = f(t, u(t)) \quad \text{in } X^*.$$

Double-phase operator: verification of (S0) – (S2)

The continuity of $p(\cdot)$, $p(0) = p(T)$, and

$$S(t, 0) = 0, \quad S(t, v) = 0 \implies v = 0,$$

give (S0).

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Since both phases are monotone,

$$\langle S(t, v) - S(t, w), v - w \rangle \geq C_\beta \|v - w\|_X^\beta,$$

while

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Moreover,

$$\|S(t, v)\|_{X^*} \leq C(1 + \|v\|_X^{\beta-1}),$$

uniformly in t , so (S2) follows.

A nonlocal $p(t)$ -Laplacian-type problem

$$\begin{aligned} \frac{\partial}{\partial t} \left[\|u_t(t)\|_{H_0^1(\Omega)}^{p(t)-2} (-\Delta u_t) \right] &= f(t, u) \quad \text{in } (0, T) \times \Omega, \\ u(t, \cdot) &= 0 \quad \text{on } \partial\Omega, \\ u(0, \cdot) &= u(T, \cdot), \quad u_t(0, \cdot) = u_t(T, \cdot). \end{aligned}$$

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$$X = H_0^1(\Omega), \quad \langle Ax, \phi \rangle = \int_{\Omega} \nabla x \cdot \nabla \phi \, dx,$$

so $A : X \rightarrow X^*$ is the Riesz map associated with $-\Delta$.

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Define $S(t, x) = \|x\|_X^{p(t)-2} Ax$, which is equivalent to

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Therefore

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Continuity, $S(t, 0) = 0$, and $p(0) = p(T)$ give (H0).

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On bounded sets,

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Finally,

$$\|S(t, x)\|_{X^*} = \|x\|_X^{p(t)-1},$$

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Pseudomonotone nonlinearities

A map $F : X \rightarrow X^*$ is **pseudomonotone** if

$$x_n \rightharpoonup x, \quad \limsup_{n \rightarrow \infty} \langle F(x_n), x_n - x \rangle \leq 0$$

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Relation with our assumptions

If

$$f = m + k,$$

where m is monotone and hemicontinuous and k is sequentially weak-to-strong continuous, then f is pseudomonotone.

Possible extension of the Galerkin approach

The decomposition

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Under suitable growth assumptions, one expects the associated Nemytskii operator on the periodic Sobolev space to be pseudomonotone, allowing the same weak-limit passage.

Hence the Hartman–Galerkin strategy should extend to

pseudomonotone f + Hartman-type a priori bounds.